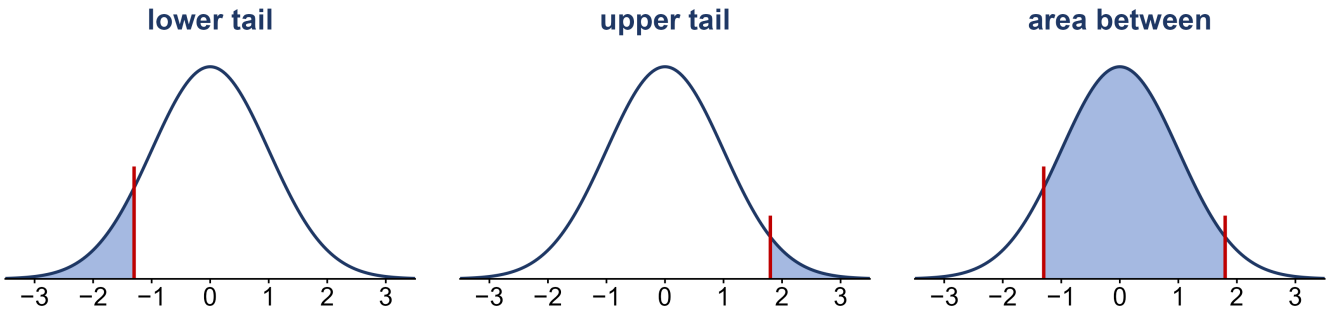
Week 9 Notes

STAT 251 SECTION 01

Finding normal probabilities

In Week 7 we said that for a continuous random variable, probability is **area under the density curve**, and that a single value has probability zero, $P(X = c) = 0$. So the questions we ask about a continuous variable are always about an *interval*: below some value, above it, or between two values. For the normal distribution those areas come from the **standard normal table** (the z-table), which we first used in Week 6.

The catch is that the table gives only **one** kind of area: the area to the **left** of z, $P(Z \le z)$, where $Z \sim N(0, 1)$ is the standard normal. Every other area has to be built out of that one.



- **Lower tail.** $P(Z < z)$ is the table value itself.
- **Upper tail.** The whole area under the curve is 1, so the area to the right is what is left over:

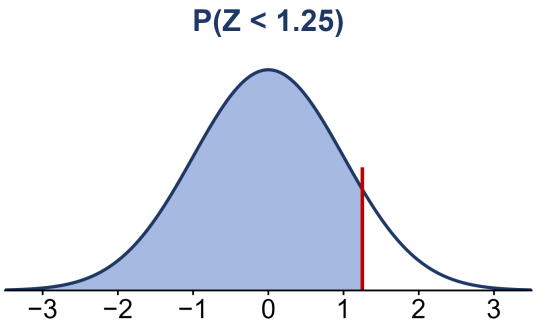
$$P(Z > z) = 1 - P(Z \le z)$$

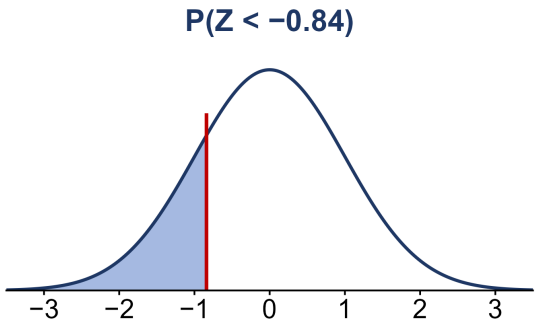
- **Area between.** Take everything to the left of the larger value, and remove everything to the left of the smaller one:

$$P(a < Z < b) = P(Z \le b) - P(Z \le a)$$

Because $P(Z = z) = 0$, it does not matter whether the inequality is $<$ or $\le$. Always **sketch the curve and shade the area first**: the picture tells you which rule to use, and whether your answer should be big or small.

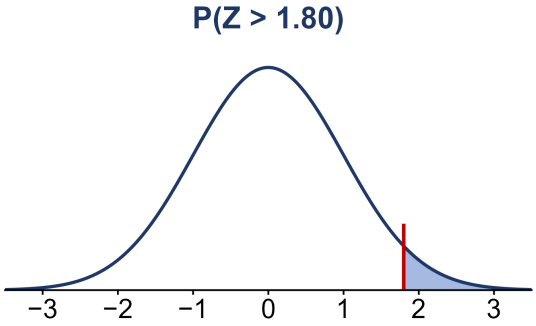
The lower tail

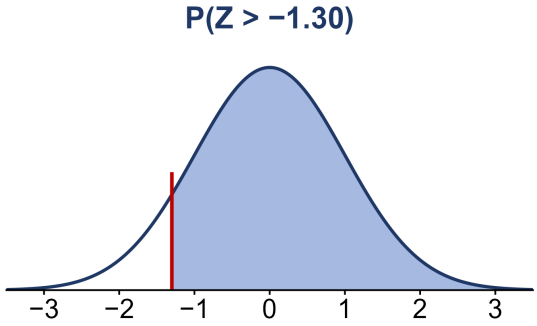




- $P(Z < 1.25)$: row 1.2, column .05 of the table gives **0.8944**.
- $P(Z < -0.84)$: row -0.8, column .04 gives **0.2005**. A negative z is below the mean, so the area to its left is less than one half.

The upper tail

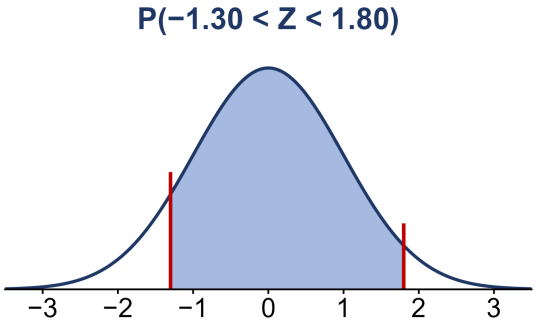


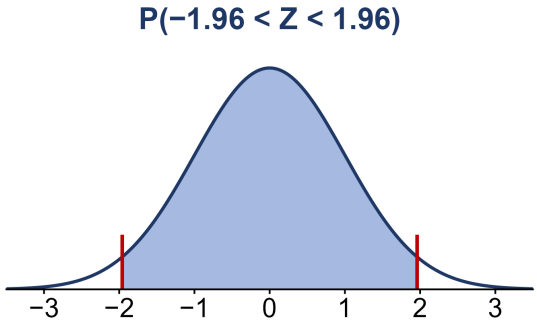


- $P(Z > 1.80) = 1 - P(Z \le 1.80) = 1 - 0.9641 = 0.0359$.
- $P(Z > -1.30) = 1 - P(Z \le -1.30) = 1 - 0.0968 = 0.9032$.

The normal curve is symmetric, so the area to the right of -1.30 is the same as the area to the left of $+1.30$: the table gives $P(Z \le 1.30) = 0.9032$, the same answer.

The area between





- $P(-1.30 < Z < 1.80) = 0.9641 - 0.0968 = 0.8673$.
- $P(-1.96 < Z < 1.96) = 0.9750 - 0.0250 = 0.9500$.

Remember that second one. Exactly 95% of a normal distribution lies within **1.96** standard deviations of the mean; the empirical rule's "2" is that number rounded. It is where the 2 in a 95% confidence interval comes from.

Practice

For $Z \sim N(0, 1)$, sketch and shade each area, then use the table:

a. $P(Z \leq 2.00)$ (b) $P(Z > 0.50)$ (c) $P(-1 < Z < 1)$ (d) $P(-2 < Z < 2)$

Answers:

- a. 0.9772
- b. $1 - 0.6915 = 0.3085$
- c. $0.8413 - 0.1587 = 0.6826$, about 68%
- d. $0.9772 - 0.0228 = 0.9544$, about 95%

c. and (d) are the empirical rule from Week 6: about 68% of a normal distribution lies within one standard deviation of the mean and about 95% within two.

Any normal distribution

The same three rules work for any normal variable $X \sim N(\mu, \sigma)$: first convert each value to a z-score, $z = (x - \mu)/\sigma$, then find the area for Z. For example, if $X \sim N(5, 10)$, then $P(X \geq 25) = P(Z \geq 2) = 1 - 0.9772 = 0.0228$. The central limit theorem, next, uses exactly this, with the standard error in place of σ.

The central limit theorem and an interval for a mean

This section is a summary. The full version, with the simulations, pictures and worked examples, is on the Week 8 notes page, starting at "The central limit theorem" (lecturemenu8.html#the-central-limit-theorem).

The theorem

In Week 8 we found the **center** and **spread** of the sampling distribution of $\bar{x}$: mean μ and standard error $\sigma/\sqrt{n}$. The **central limit theorem** (CLT) gives its **shape**:

- **Central limit theorem.** Take random samples of size n from **any** population with mean μ and standard deviation σ. If n is large enough, the sampling distribution of $\bar{x}$ is approximately normal:

$$\bar{x} \sim N\left(\mu,\ \frac{\sigma}{\sqrt{n}}\right)$$

  A sample proportion is a sample mean of 0s and 1s, so the same is true of $\hat{p}$:

$$\hat{p} \sim N\left(p,\ \sqrt{\frac{p(1-p)}{n}}\right)$$

The surprising part is "**any** population". Even when the population is skewed, like the minutes until the next goal at the World Cup, averages of enough observations are bell shaped. The larger n is, the better the approximation. The textbook's guides for "large enough" are:

- $n \geq 30$ for $\bar{x}$. A population that is close to normal needs less; one that is very skewed or has extreme outliers may need more.
- $np > 5$ and $n(1-p) > 5$ for $\hat{p}$, so there are enough successes *and* failures.

Using it

Because $\bar{x}$ is approximately normal, we can find probabilities about it with the z-table, exactly as in the first part of these notes. The one change is that we standardize with the **standard error**, not σ:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

For example, a World Cup's average over $n = 64$ matches has $\mu = 2.69$ goals and standard error 0.205, so $P(\bar{x} \geq 3) = P(Z \geq 1.51) = 1 - 0.9345 = 0.0655$.

In particular, about 95% of the time a statistic lands within **2 standard errors** of its parameter:

$$P\left(\mu - 2\frac{\sigma}{\sqrt{n}}\ \leq\ \bar{x}\ \leq\ \mu + 2\frac{\sigma}{\sqrt{n}}\right) \approx 0.95$$

A 95% confidence interval for a mean

If $\bar{x}$ is within 2 standard errors of μ, then μ is within 2 standard errors of $\bar{x}$. Turning the statement around gives the **95% confidence interval** for μ:

$$\bar{x}\ \pm\ 2\frac{\sigma}{\sqrt{n}}$$

which has the form **point estimate ± margin of error**. For example, a sample of $n = 20$ songs with $\bar{x} = 3.64$ minutes and $\sigma = 1.51$ gives $3.64 \pm 2(0.34)$, from **2.96 to 4.32 minutes**: "We estimate with 95% confidence that the mean length of all 18,860 songs is between 2.96 and 4.32 minutes."

The 95% describes the **method**, not this one interval. About 95% of samples give an interval that captures μ. Once an interval is computed, it either contains μ or it does not. It is **not** correct to say there is a 95% probability that μ is between 2.96 and 4.32.

A confidence interval for a proportion

The interval for a mean was point estimate ± 2 standard errors. The same idea works for a proportion. By the CLT, for large n the sample proportion $\hat{p}$ is approximately normal with mean p and standard error $\sqrt{p(1-p)/n}$, so about 95% of samples give a $\hat{p}$ within 2 standard errors of p. Turning that around, as we did for the mean:

$$\hat{p} \pm 2\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

is a 95% confidence interval for p. There is one change from the formula for the standard error: we do not know p (that is what we are estimating), so the standard error uses $\hat{p}$ in its place.

When it works. The interval leans on the CLT, so check that there are enough successes **and** failures in the sample: $n\hat{p} > 5$ and $n(1-\hat{p}) > 5$.

The data: teens and cell phones

In Week 1 we met a Pew Research Center survey that asked 743 U.S. teens aged 13 to 17, "Are you losing focus in class by checking your cell phone?" The published counts are:

Response	Teens
Never	289
Rarely	216
Sometimes	178
Often	60
Total	743

Call a teen a "success" if they answered "sometimes" or "often":

- X = 1 if a teen loses focus in class sometimes or often, and 0 otherwise.

Example: losing focus in class

Estimate p, the proportion of **all** U.S. teens who lose focus in class sometimes or often.

- **Point estimate.** $178 + 60 = 238$ of the $n = 743$ teens, so $\hat{p} = 238/743 \approx 0.320$.
- **Check.** There are 238 successes and 505 failures, both far more than 5.
- **Standard error.** $\sqrt{0.320 \times 0.680/743} \approx 0.017$.
- **Interval.** $0.320 \pm 2(0.017) = 0.320 \pm 0.034$, from **0.286 to 0.354**.

We would report it as: **"We estimate with 95% confidence that between 28.6% and 35.4% of U.S. teens lose focus in class sometimes or often by checking their phones."** As with the mean, the 95% describes the method: about 95% of samples of 743 teens would give an interval that captures p.

Confidence levels and critical values

Above we introduced the **confidence interval** which is used to estimate a parameter with an interval of values. We can choose how wide to make the confidence interval by selecting the **confidence level** - which is the probability that the interval will contain the true parameter value **before** the data are gathered. The confidence level is controlled via the standard score z. The standard score gives the number of standard errors to add and subtract from our estimate. Therefore, we must look up the appropriate standard score z from the standard normal table.

We saw that for a 95% confidence interval, we must add and subtract approximately $z = 2$ standard errors from our estimate $\bar{x}$ or $\hat{p}$. z is sometimes referred to as the **critical value**. The critical value denotes the upper and lower bounds of the confidence interval. In general, if we wish to compute the $100(1-\alpha)\%$ confidence interval for the statistics $\bar{x}$ or $\hat{p}$, then we must add and subtract $z_{1-\alpha/2}$ standard errors. The quantity $\alpha/2$ denotes the area that is divided among the two tails of the standard normal distribution. For example, for a 95% confidence interval $\alpha = 0.05$ and $\alpha/2 = 0.025$.

For example, for a 90% confidence $\alpha = 0.1$, and the corresponding critical value is $z_{1-0.1/2} = z_{0.95}$ which is the 95th percentile from the standard normal distribution. The table below gives several confidence levels, the corresponding values of α and critical values.

Confidence Level	α	critical value: $z_{1-\alpha/2}$
80%	0.20	1.28
85%	0.15	1.44

Confidence Level		critical value:
90%	0.10	1.64
95%	0.05	1.96
99%	0.01	2.58

- Try it out: give the confidence level and critical value for the $100(1-\alpha)\%$ confidence intervals with $\alpha = 0.25$, $\alpha = 0.08$, and $\alpha = 0.03$.

An alternative confidence interval for a population mean.

Recall the confidence interval that we introduced for estimating the population mean:

$$\bar{x} \pm z\frac{\sigma}{\sqrt{n}}$$

The confidence interval above is based on the assumption that the population standard deviation σ is known. When σ is known the sampling distribution of the $\bar{x}$ is approximately $N(\mu, \sigma/\sqrt{n})$. However, we rarely know the value of σ so we must approximate its value with the sample standard deviation s to get

$$\bar{x} \pm z\frac{s}{\sqrt{n}}$$

Unfortunately, the confidence level of the confidence interval above will tend to be less than specified. As a result, more confidence intervals will fail to cover the true parameter value than expected.

- Consider the following example: A social scientist is conducting a study to investigate the average time spent daily on social media by college students. Each week the scientist surveys five college students and asks them to report their time spent on social media. For each sample, the scientist computes the 95% confidence interval for the mean time the students spent on social media. They repeat this process 100 times. Suppose the population mean time spent on social media is $\mu = 120$ minutes with a standard deviation of $\sigma = 10$ minutes. The plot below shows 100 confidence intervals computed using $\bar{x} \pm z\frac{s}{\sqrt{n}}$ where each CI is computed from a sample of 5 observations from $N(120, 10)$:

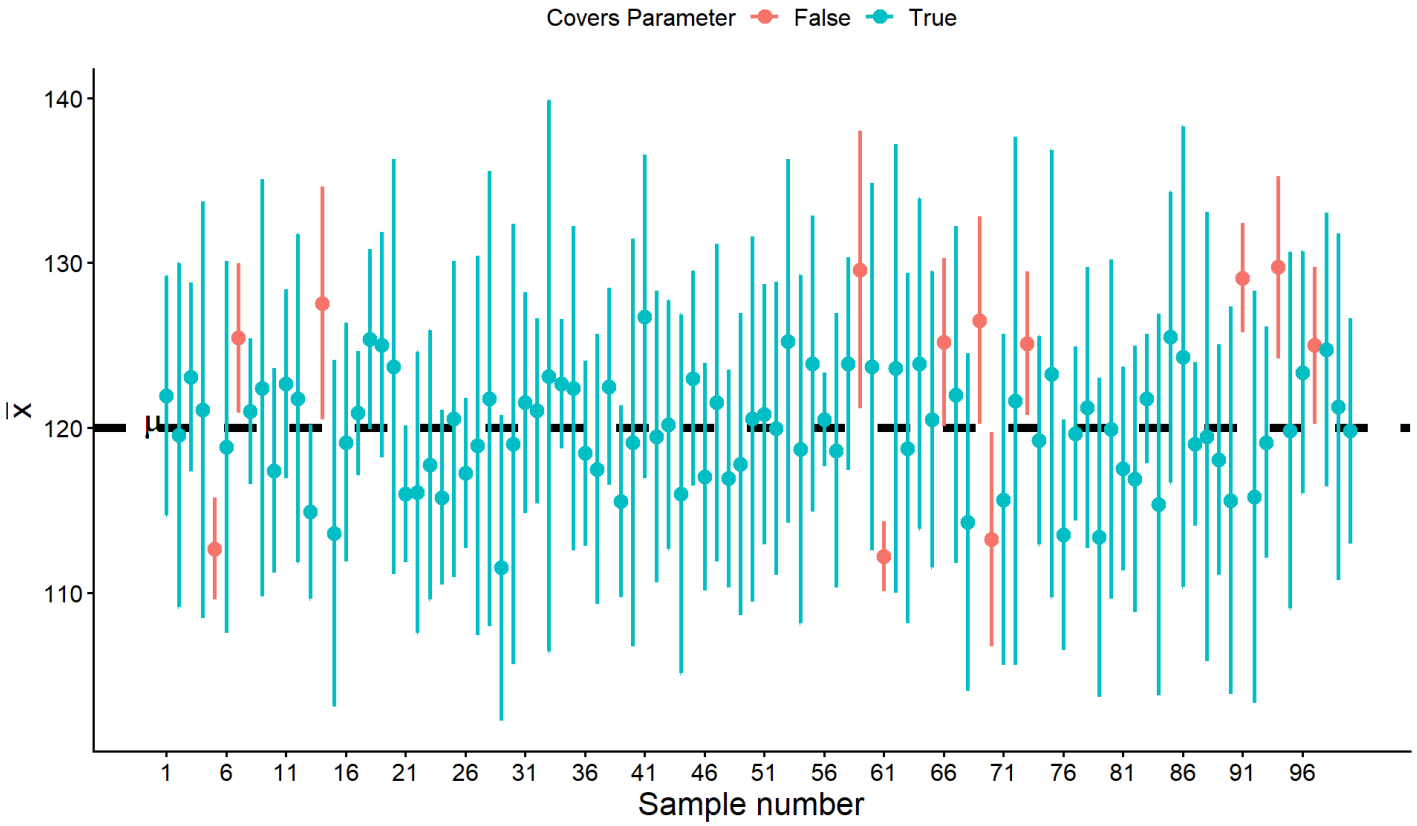
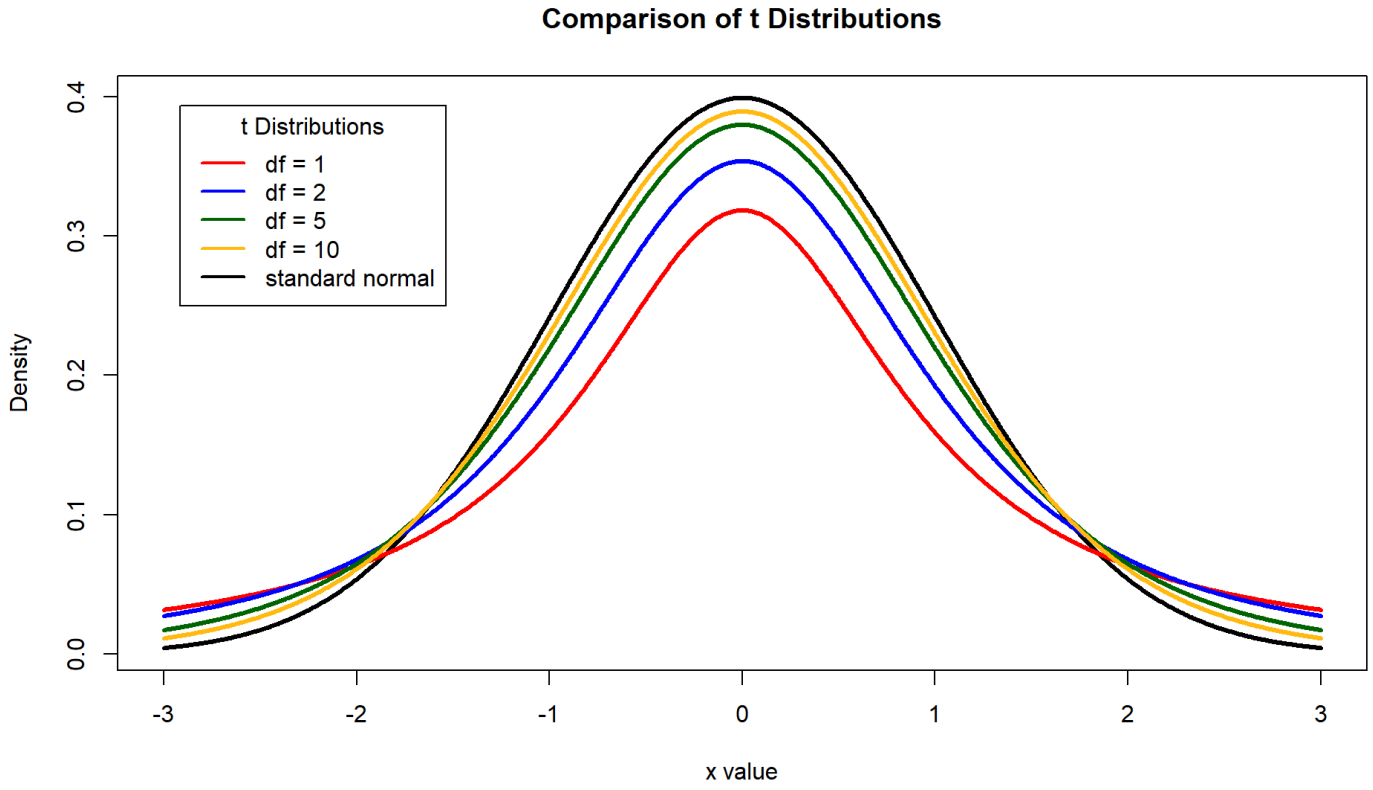


Figure: Each point represents the sample mean from a hypothetical sample of 5 college students. The whiskers extending from each point give the coverage of the confidence interval computed for each sample. The black dotted line gives the value of hypothetical population mean time spent on social media. The CI's highlighted in red represent those that fail to capture the population mean

Based on the plot and example above, the actual proportion of confidence intervals that fail to capture the population mean is 12%, which is much higher than the expected frequency of only 5%. This occurs because when we substitute $s/\sqrt{n}$ for $\sigma/\sqrt{n}$ the standardized statistic $\frac{\bar{x}-\mu}{s/\sqrt{n}}$ no longer follows a standard normal distribution. Instead, it follows a distribution called the t**-distribution**.

Like the normal distribution, the t distribution is a bell curve but it has "heavier tails" - that is to say, it has more probability density in the tails of the distribution.



The amount of probability density in the tails of a t distribution is controlled by the **degrees of freedom** of the distribution. The confidence interval for μ that uses $s/\sqrt{n}$ instead of $\sigma/\sqrt{n}$ is given by

$$\bar{x} \pm t\frac{s}{\sqrt{n}}$$

The quantity t is called the t-score and is the $1-\alpha/2$ percentile of t-distribution with $n-1$ degrees of freedom denoted $t(n-1)$. Like the standard score z, we must look up the t-score corresponding to the desired confidence level from a t-distribution table:

Upper tail	0.25	0.2	0.15	0.125	0.1	0.075	0.05	0.025	0.005	0.0025	5e-04
Two tail	0.5	0.4	0.3	0.25	0.2	0.15	0.1	0.05	0.01	0.005	0.001
DF											
1	1	1.376	1.963	2.414	3.078	4.165	6.314	12.706	63.657	127.321	636.619
2	0.816	1.061	1.386	1.604	1.886	2.282	2.92	4.303	9.925	14.089	31.599
3	0.765	0.978	1.25	1.423	1.638	1.924	2.353	3.182	5.841	7.453	12.924
4	0.741	0.941	1.19	1.344	1.533	1.778	2.132	2.776	4.604	5.598	8.61
5	0.727	0.92	1.156	1.301	1.476	1.699	2.015	2.571	4.032	4.773	6.869

Upper tail	0.25	0.2	0.15	0.125	0.1	0.075	0.05	0.025	0.005	0.0025	5e-04
6	0.718	0.906	1.134	1.273	1.44	1.65	1.943	2.447	3.707	4.317	5.959
7	0.711	0.896	1.119	1.254	1.415	1.617	1.895	2.365	3.499	4.029	5.408
8	0.706	0.889	1.108	1.24	1.397	1.592	1.86	2.306	3.355	3.833	5.041
9	0.703	0.883	1.1	1.23	1.383	1.574	1.833	2.262	3.25	3.69	4.781
10	0.7	0.879	1.093	1.221	1.372	1.559	1.812	2.228	3.169	3.581	4.587
11	0.697	0.876	1.088	1.214	1.363	1.548	1.796	2.201	3.106	3.497	4.437
12	0.695	0.873	1.083	1.209	1.356	1.538	1.782	2.179	3.055	3.428	4.318
13	0.694	0.87	1.079	1.204	1.35	1.53	1.771	2.16	3.012	3.372	4.221
14	0.692	0.868	1.076	1.2	1.345	1.523	1.761	2.145	2.977	3.326	4.14
15	0.691	0.866	1.074	1.197	1.341	1.517	1.753	2.131	2.947	3.286	4.073
16	0.69	0.865	1.071	1.194	1.337	1.512	1.746	2.12	2.921	3.252	4.015
17	0.689	0.863	1.069	1.191	1.333	1.508	1.74	2.11	2.898	3.222	3.965
18	0.688	0.862	1.067	1.189	1.33	1.504	1.734	2.101	2.878	3.197	3.922
19	0.688	0.861	1.066	1.187	1.328	1.5	1.729	2.093	2.861	3.174	3.883
20	0.687	0.86	1.064	1.185	1.325	1.497	1.725	2.086	2.845	3.153	3.85
21	0.686	0.859	1.063	1.183	1.323	1.494	1.721	2.08	2.831	3.135	3.819
22	0.686	0.858	1.061	1.182	1.321	1.492	1.717	2.074	2.819	3.119	3.792
23	0.685	0.858	1.06	1.18	1.319	1.489	1.714	2.069	2.807	3.104	3.768
24	0.685	0.857	1.059	1.179	1.318	1.487	1.711	2.064	2.797	3.091	3.745
25	0.684	0.856	1.058	1.178	1.316	1.485	1.708	2.06	2.787	3.078	3.725
26	0.684	0.856	1.058	1.177	1.315	1.483	1.706	2.056	2.779	3.067	3.707
27	0.684	0.855	1.057	1.176	1.314	1.482	1.703	2.052	2.771	3.057	3.69
28	0.683	0.855	1.056	1.175	1.313	1.48	1.701	2.048	2.763	3.047	3.674
29	0.683	0.854	1.055	1.174	1.311	1.479	1.699	2.045	2.756	3.038	3.659

Upper tail	0.25	0.2	0.15	0.125	0.1	0.075	0.05	0.025	0.005	0.0025	5e-04
30	0.683	0.854	1.055	1.173	1.31	1.477	1.697	2.042	2.75	3.03	3.646
40	0.681	0.851	1.05	1.167	1.303	1.468	1.684	2.021	2.704	2.971	3.551
50	0.679	0.849	1.047	1.164	1.299	1.462	1.676	2.009	2.678	2.937	3.496
100	0.677	0.845	1.042	1.157	1.29	1.451	1.66	1.984	2.626	2.871	3.39
Z-score	0.674	0.842	1.036	1.15	1.282	1.44	1.645	1.96	2.576	2.807	3.291
Confidence Level:	50%	60%	70%	75%	80%	85%	90%	95%	99%	99.5%	99.9%

The table for the t distribution is organized a little bit differently than the one we've seen previously for the standard normal distribution. The left-most column gives the degrees of freedom, columns give the different upper tail and two tail probabilities, and the values in the middle of the table are the t scores that correspond to those probabilities.

Use the table above to verify that the following t-scores corresponding to each confidence level and sample size are correct.

Confidence Level	Sample size	α	critical value: $t_{1-\alpha/2}$
90%	$n = 3$	0.10	2.92
85%	$n = 5$	0.15	1.78
95%	$n = 10$	0.05	2.26
99%	$n = 14$	0.01	3.01
80%	$n = 22$	0.20	1.32

One nice property of the t-distribution is that it converges to the standard normal distribution as n goes to infinity. Stated another way, a t-distribution with ∞ degrees of freedom is exactly the standard normal distribution. This means that with large enough sample size, both the confidence interval based on the standard normal distribution and the one based on the t distribution should yield nearly identical results. In fact, for $n \geq 30$ the difference between the standard normal distribution and t-distribution is nearly indistinguishable.